

Sheet 1

File: io-v2.0-sheet1.kicad_sch

Sheet 2

File: io-v2.0-sheet2.kicad_sch

Sheet 3

File: io-v2.0-sheet3.kicad_sch

Sheet 4

File: io-v2.0-sheet4.kicad_sch

Sheet 5

File: io-v2.0-sheet5.kicad_sch

Sheet 6

File: io-v2.0-sheet6.kicad_sch

Sheet 7

File: io-v2.0-sheet7.kicad_sch

Circuit source: io_v2_netlist.py; generated by gen_io_v2.py; see README.md
Sheets 1-6 = I/O card v1.1 (unchanged); sheet 7 = CompactFlash on P4/P5 (from the CF card v1.0)

Sheet: /
File: io-v2.0.kicad_sch

Title: YACC1 I/O card v2.0

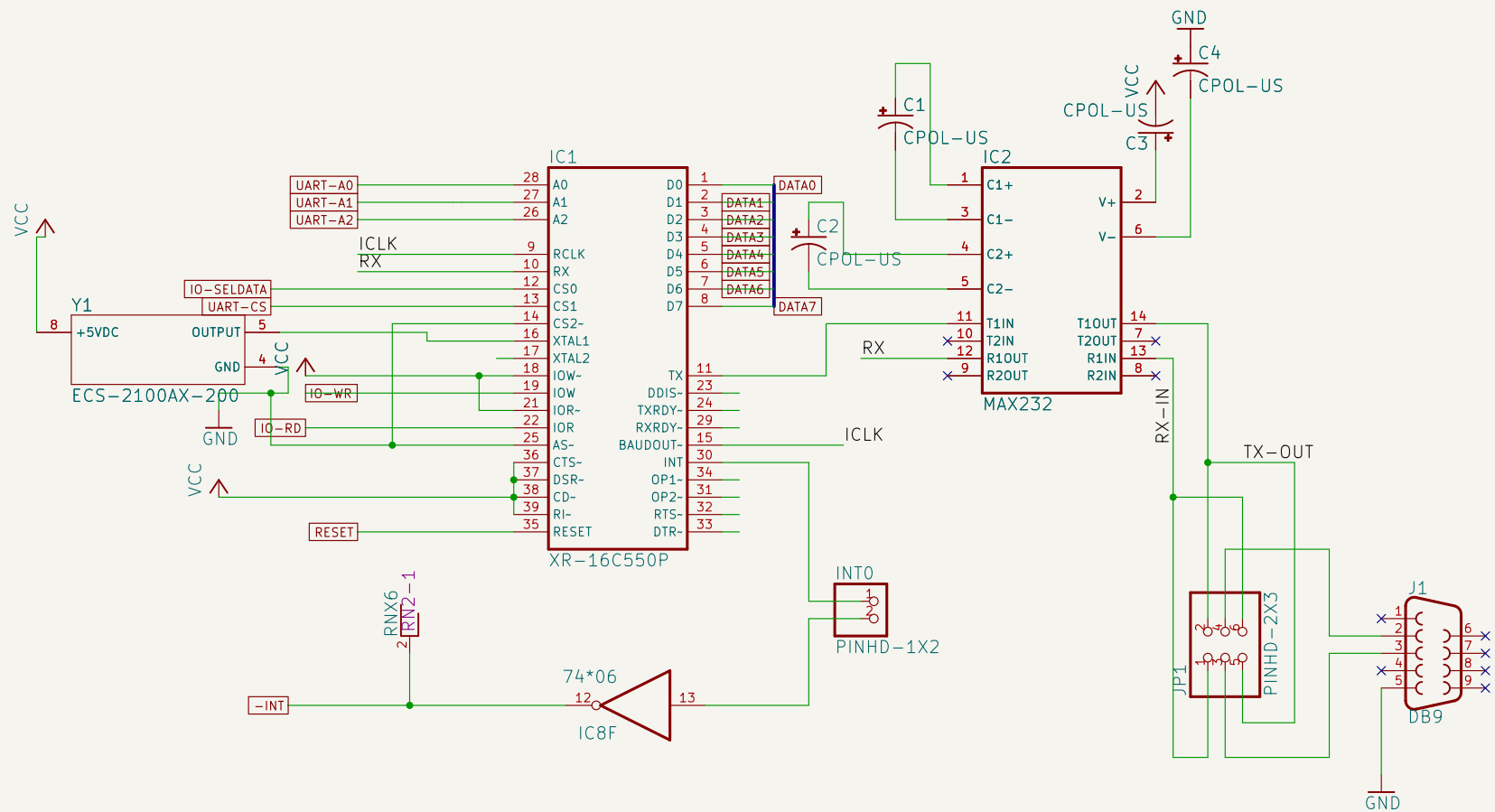
Size: A4Date: 2026-09-23

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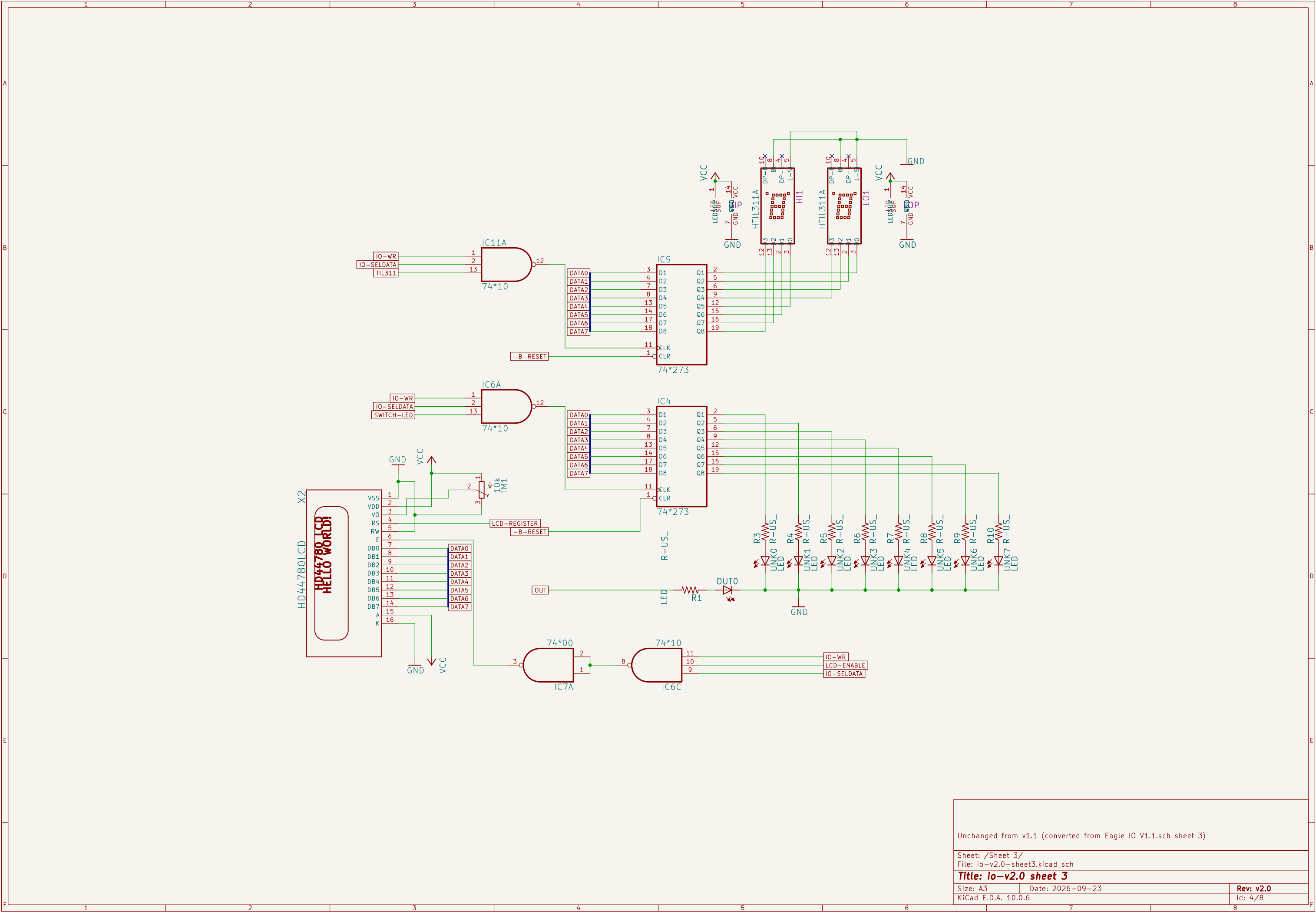
Unchanged from v1.1 (converted from Eagle IO V1.1.sch sheet 2)

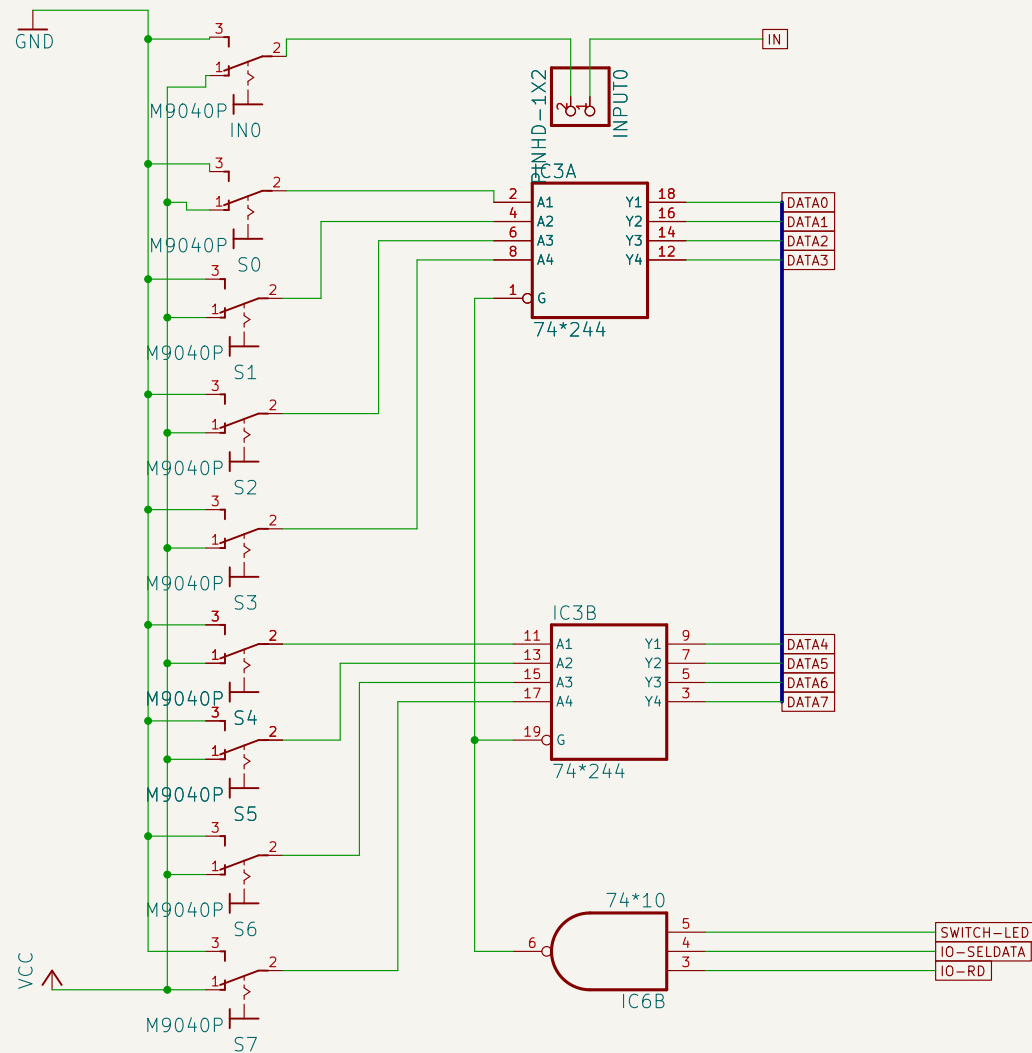
Sheet: /Sheet 2/
File: io-v2.0-sheet2.kicad_sch

Title: io-v2.0 sheet 2

Size: A4 Date: 2026-09-23
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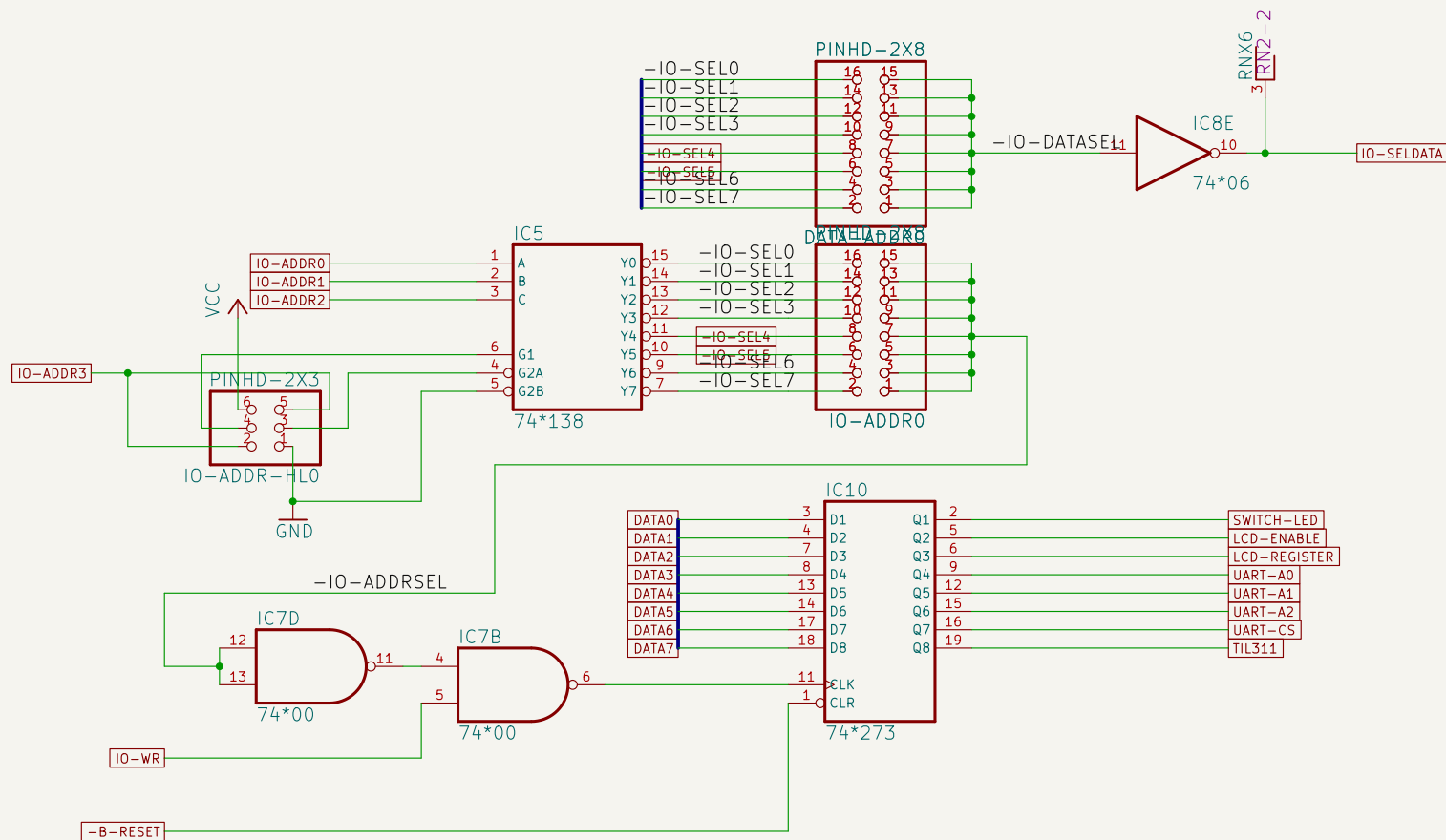
Unchanged from v1.1 (converted from Eagle IO V1.1.sch sheet 4)

Sheet: /Sheet 4/
File: io-v2.0-sheet4.kicad_sch

Title: io-v2.0 sheet 4

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I/O card v2.0: the IO-ADDR-HL strap MUST decode P0-P7 (jumpers 3-5 and 4-6):
 IC5 Y4 (-IO-SEL4) = P4 = CF register-select latch, Y5 (-IO-SEL5) = P5 = CF data (sheet 7).
 The IO-ADDR and DATA-ADDR jumpers must not select P4 or P5 (pins 7-8, 5-6).

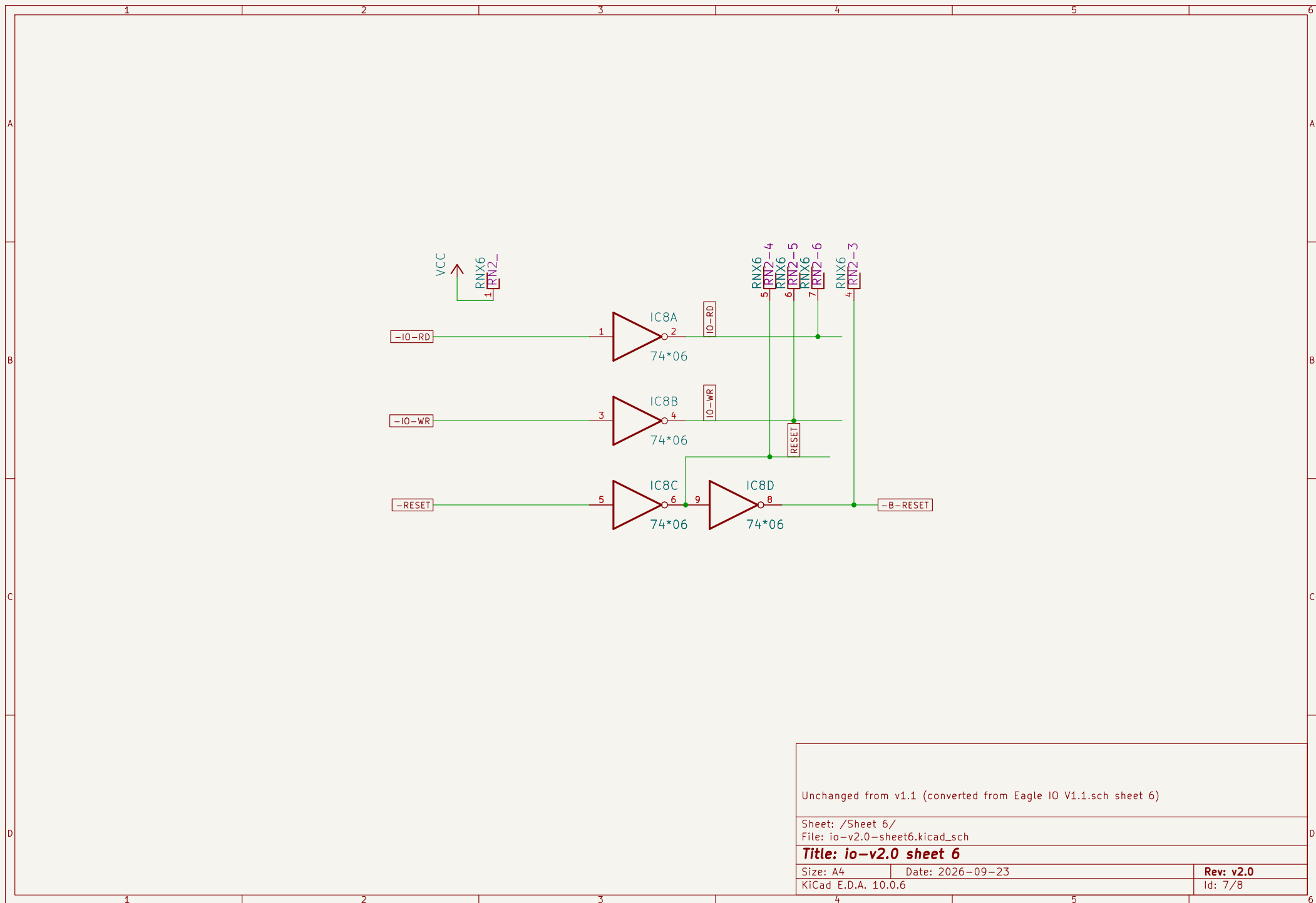
v1.1 sheet 5 + -IO-SEL4/-IO-SEL5 made global (CF section, sheet 7) + strap note

Sheet: /Sheet 5/
 File: io-v2.0-sheet5.kicad_sch

Title: io-v2.0 sheet 5

Size: A4 Date: 2026-09-23
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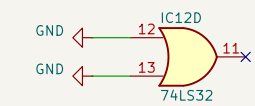
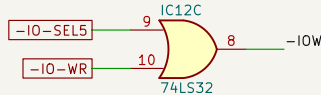
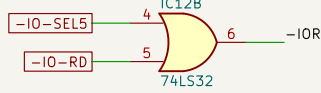
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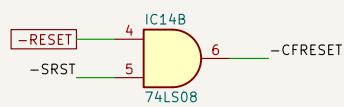
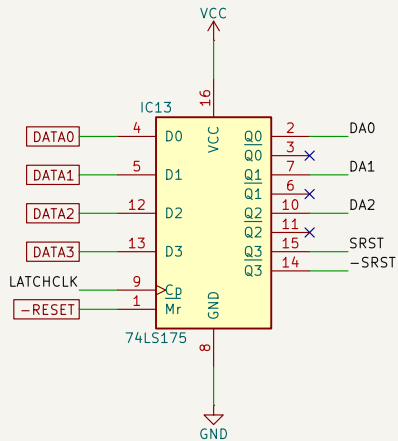
P4 / P5 STROBE GATING (IC12 74LS32)

–IO–SEL4 = IC5 Y4 (sheet 5): port P4
–IO–SEL5 = IC5 Y5 (sheet 5): port P5
IC5 decodes P0–P7 only with the
IO–ADDR–HL strap in P0–P7.

gate 1: P4 latch clock
= –IO–SEL4 OR –IO–WR
gate 2: CF –IOR
= –IO–SEL5 OR –IO–RD
gate 3: CF –IOW
= –IO–SEL5 OR –IO–WR
gate 4: unused

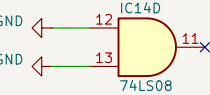
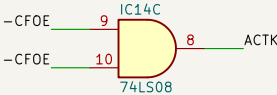
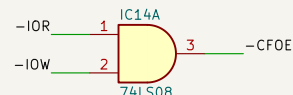
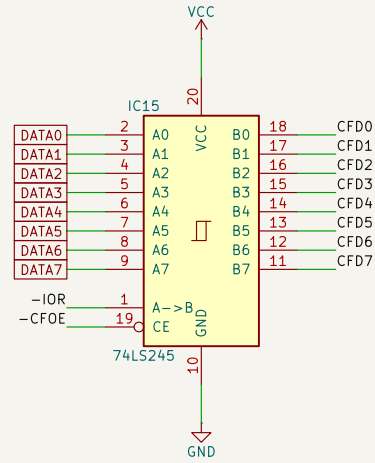


P4 LATCH: DA0–2 + CF RESET (IC13 74LS175)



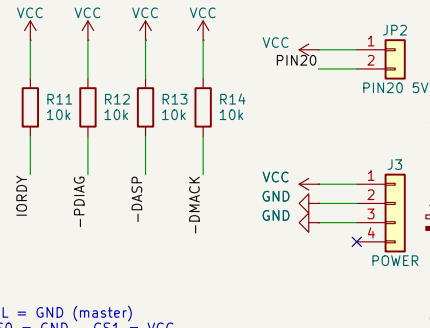
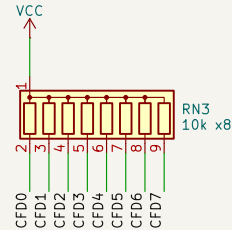
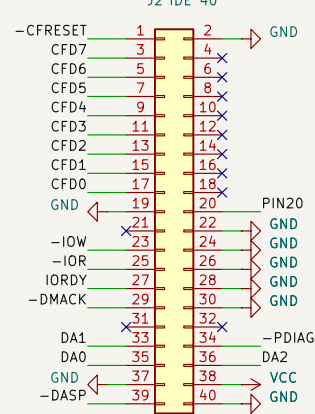
IC14 gate 2: –CFRESET =
–RESET AND –SRST
(bus reset or latch bit 3)

P5 DATA BUFFER (IC15 74LS245)



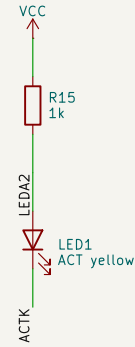
DIR = –IOR (low: CF → bus)
–CFOE = –IOR AND –IOW
ACTK sinks the ACT LED

IDE HEADER J2 (TO THE CF ADAPTER)



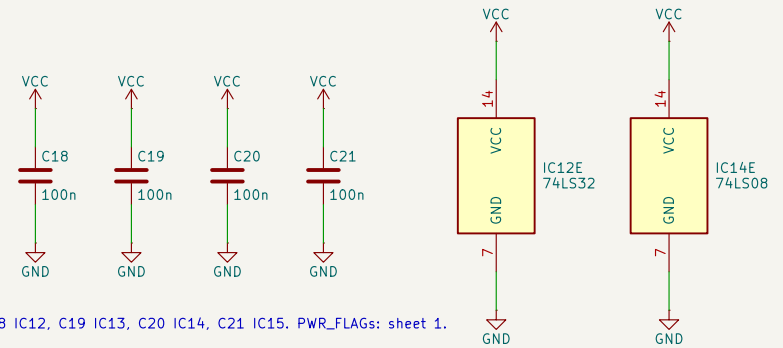
CSEL = GND (master)
–CS0 = GND, –CS1 = VCC

ACT LED



ACT: any P5 access
(PWR LED: the card's PWR0;
no DASP LED on v2.0)

POWER + DECOUPLING (one 100 nF per new IC)



DESIGN NOTES (I/O CARD v2.0, CF SECTION)

- Two I/O ports on the I/O card's own decoder IC5 (sheet 5): P4 (write) = 74LS175 latch IC13 – bits 0–2 = the ATA task–file register (CF DA0–2), bit 3 = CF reset (1 = held in reset); P5 (read/write) = the selected ATA register, 8–bit True IDE, through the 74LS245 IC15. The IO–ADDR–HL strap MUST decode P0–P7; the IO–ADDR / DATA–ADDR jumpers must not select P4 or P5 (they would put the control latch or the UART/switch port on a CF port).
- IC12 (74LS32) ORs each port select with –IO–WR / –IO–RD, so –IOR, –IOW and the latch clock are low only during the bus strobe for this port (the microcode sets IOADDR two steps before the strobe and holds it through it).
- CS0 (J2 pin 37) is tied LOW and –CS1 (pin 38) HIGH: the strobes alone define every cycle (CF card v1.0 note 3, docs/cards/cf.md section 4: gating –CS0 would break the chip–select hold time after –IOR).
- J2 = 40–pin IDE header, a short ribbon to a CF–to–IDE adapter mounted above the card on standoffs. Adapter power: J3 (1 = +5 V, 2 and 3 = GND, 4 = +12 V, not used) or, with JP2 fitted, IDE pin 20 for adapters powered there.
- Pull–ups: CF D0–7 (RN3: an empty adapter reads \$FF = 'no card' to the ROM driver), IORDY, –PDIAG, –DASP; –DMACK held inactive; CSEL grounded (master). SRST (IC13 Q4) is a one–pin net by design (a probe point).
- From hardware/cards/cf/kicad/v1.0/cf_netlist.py: U2→IC12 U3→IC13 U4→IC14 U5→IC15 J1→J2 J2→J3 JP1→JP2 RN1→RN3 R1–R5→R11–R15 LED2→LED1 C2–C6→C18–C22. Dropped: U1 (74LS138: IC5 Y4/Y5 replace –P8SEL/–P9SEL), X1 (the card's bus connector), C1, the PWR LED (LED1, R7: the card has PWR0) and the DASP LED (LED3, R6).

Theory: docs/cards/cf.md, docs/cards/io.md. Circuit source: io_v2_netlist.py (edit, then re–run build.sh).

Circuit source: io_v2_netlist.py; this sheet is generated by gen_io_v2.py
From the CF card v1.0 minus its 74LS138, bus connector, PWR + DASP LEDs
CF (True IDE, 8–bit) on I/O ports P4 (register latch) / P5 (data) – docs/cards/cf.md
YACC1

Sheet: /Sheet 7/
File: io–v2.0–sheet7.kicad_sch

Title: io–v2.0 sheet 7: CompactFlash interface

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